

PRATYUSH PADHY

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EDUCATION

University of California, Irvine

Irvine, CA

B.S., Computer Science, Specialization in Intelligent Systems — GPA: 3.86/4.00

Expected Graduation: 2028

- Dean's Honor List: Fall '25, Winter '26, Spring '26 | **Organizations:** Data@UCI, AI@UCI, Cyber@UCI, Hack@UCI

PROFESSIONAL EXPERIENCE

Tata Consultancy Services (TCS)

CMU Campus, Pittsburgh, PA

AI/ML Intern — TCS Medical Robotics Center

Jun. 2026 – Present

- Benchmarked task-specific **RL** against a generalist **vision-language-action** policy (**SmolVLA** 450M, **LeRobot**) across **250+** episodes on an **AgileX Piper 6-DOF** arm — the **pixel-only VLA** matched **state-based RL (85.3% vs. 81.5%)**.
- Extended **FluidLab** (ICLR '23 MPM fluid simulator) into a robotics platform: **URDF forward-kinematics** joint control (**0.0mm** error), analytic **SDF** liquid vessels on a **128³** voxel grid, an offscreen renderer, and **3** upstream engine-bug fixes.
- Diagnosed a **sparse-reward exploration collapse** and solved it by **action-space reparameterization** — no reward shaping, no demonstrations — training **PPO (Stable-Baselines3)** to **81.5% ± 3.3** liquid transfer across 3 seeds.
- Fine-tuned **SmolVLA** via two-stage **imitation learning** (real-robot pretraining → **120** PPO-generated demos) in **13 min**; budget-matched PPO trained on identical pixels scored **0%** — an **ablation** isolating pretrained vision + imitation.

Ready Tutor

Irvine, CA

Software Developer Intern

Mar. 2026 – Jun. 2026

- Designed a centralized storage layer — a storage-agnostic schema class over **MongoDB/SQLite** behind a **React** useContext provider — replacing **10+** scattered storage instances and batching persistence into a single write per user action.
- Built a course catalog with real-time retrieval of **1,200+** UCI courses — the headline feature of the funding demo that secured **\$5,500** — and resolved **30+** production bugs, including a credits-calculation error affecting schedule planning.
- Redigned the year planner from a rigid 4-year structure to support unlimited years with per-quarter course blocks.

Robotics for All

Palo Alto, CA

Software Intern

Jun. 2023 – Aug. 2024

- Deployed a comprehensive asynchronous training system with structured readiness assessments for **24+** instructors, standardizing the onboarding pipeline and ensuring consistent teaching quality across all program locations.
- Coached coding classes of **20** students, designed a parent feedback system to continuously improve curriculum quality, and interviewed volunteers to assess teaching readiness; awarded the Presidential Volunteer Service Award (Gold, **250+** hours).

PROJECTS

Chronicle | Python, FastAPI, Postgres/pgvector, Next.js | chronicles-weld.vercel.app

Jun. 2026 – Present

- Built a job aggregation platform ingesting **~26K** open roles from **600+** companies' ATS boards (**Greenhouse, Lever, Ashby**) via async fan-out with per-company fault isolation and idempotent **24–48h** refreshes that dedupe cross-source postings.
- Engineered hybrid search fusing **Postgres full-text (ts_rank_cd, GIN index)** with semantic ranking (int8 **ONNX** MiniLM, **pgvector** HNSW) via **RRF** at **p95 19ms** on a 20K-job corpus (local bench); a profile-vector recommender lifted **NDCG@10** 0.58→0.81, **recall@50** 0.71→0.96, and **MRR** 0.80→0.98 over the rule baseline (n=24 personas, bootstrap 95% CIs).
- Shipped **Google OAuth** accounts, kanban application tracking, saved-search email alerts, and a **Manifest V3** browser extension that instantly autofills **Greenhouse/Lever/Ashby** job applications from a stored user profile.

Forge | Python, PyTorch, Unsloth, TRL, vLLM, Ollama | forge-grpo.vercel.app

Jul. 2026

- Post-trained **Qwen2.5-1.5B** to reason via **GRPO** — RL from **verifiable rewards**, no human labels or reward model — with **Unsloth + TRL**, lifting **GSM8K pass@1** 58.8%→**70.0%** on 1,319 held-out problems in **86 min** at **3.64 GiB** peak VRAM.
- Designed the reward stack (correctness, format, graded tag-presence), diagnosing a zero-reward cold start that nulled GRPO's gradient; verified **100%** output-format compliance and no catastrophic forgetting on **ARC-Challenge** (69.5%→68.5%).
- Quantized to **GGUF Q4_K_M (3.15× smaller)** and served at **228 tok/s (Ollama)** and **674 tok/s** batched (**vLLM** multi-LoRA) behind a live base-vs-tuned demo — **RL across domains: PPO** for manipulation, **GRPO** for reasoning.

Lumina | Python, FastAPI, Postgres/pgvector, React | lumina-rag.vercel.app

May – Jul. 2026

- Built a production-shaped multimodal **RAG** service (PDFs, images, text) with hybrid retrieval — true **Okapi BM25** fused with dense **MiniLM/pgvector** HNSW via **RRF** — plus a **cross-encoder reranker**, **Gemini** vision ingestion, **SSE** streaming, rate limiting, and OpenAI-compatible **provider routing (Groq/Ollama/vLLM → Gemini fallback)**.
- Measured every retrieval component on a frozen **52-question** eval with hand-verified ground truth: per-component **ablations** lifted **NDCG@10** 0.47→0.79 and **recall@5** 0.57→0.87 over the dense-only baseline, with per-query **pipeline tracing**.
- Calibrated a two-stage **confidence-refusal gate** (cross-encoder threshold + bi-encoder second chance): **zero false answers, zero false refusals** across the full threshold sweep — the system cites sources or declines instead of hallucinating.

TECHNICAL SKILLS

Languages: Python, Java, R, SQL, HTML, CSS, JavaScript, TypeScript

ML & Concepts: reinforcement learning (PPO, GRPO), imitation learning & VLA fine-tuning, RAG & hybrid retrieval, LLM fine-tuning & quantization, embeddings & vector search, recommender systems, evaluation & ablations, NLP

Libraries & Frameworks: PyTorch, TensorFlow, scikit-learn, Transformers, Unsloth, TRL, vLLM, LeRobot, Stable-Baselines3, Gymnasium, OpenCV, NumPy, Pandas, Matplotlib, React, Next.js, FastAPI, Tailwind CSS

Tools & Platforms: Git, GitHub Actions (CI), Docker, pytest, PostgreSQL, Neon, Ollama, ROS2, Gazebo, FluidLab, WSL2/CUDA, Vercel, Render, Google Cloud Run, Hugging Face Spaces, MongoDB, SQLite